BoomBikes Shared Bike Demand Prediction

PART A - Assignment-Based Subjective Questions

Q1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

After converting the numerically encoded columns into meaningful string labels, box plots and bar charts were created for each categorical feature against **cnt**. The inferences are:

- **season:** Fall records the highest median daily rentals (~5,500), followed by Summer and Winter.
- Spring has the lowest demand (~2,600 median). This confirms a strong seasonal effect:
- comfortable temperatures in fall and summer attract more riders. In the OLS model, season_Winter and season_Summer both carry significant positive coefficients (relative to the Spring baseline), while season_Spring has a large negative effect on cnt.
- **weathersit:** Clear/partly-cloudy days drive the most rentals. Misty/cloudy conditions reduce demand moderately (~10 - 15% lower than clear days). Light Snow or Rain sharply suppresses ridership (~60 - 65% lower). Weather situation is among the strongest categorical predictors, with weathersit_LightSnowRain carrying the largest negative coefficient in the final model.
- **mnth:** Demand peaks during June - September and is lowest in January - February, mirroring the temperature and season pattern. Months like September, July, and August show significantly higher rentals than January (the dropped baseline month).
- **weekday / workingday:** Registered (commuter) users drive higher rentals on working weekdays, while casual users peak on weekends. Overall, the net impact on cnt is modest but statistically significant. workingday is retained in the final model with a small positive coefficient.
- **yr:** 2019 shows substantially higher demand than 2018 across all categories, confirming a yearover-year growth trend in bike-sharing adoption. yr is one of the top 3 predictors in the final model.

Q2. Why is it important to use drop_first=True during dummy variable creation?

When a categorical variable with k distinct levels is one-hot encoded, exactly k binary columns are produced. Because these k columns always sum to 1 for every row, one column is an exact linear combination of the remaining $k - 1$ columns. This is the dummy variable trap — a special case of perfect multicollinearity.

Using **drop_first=True** removes one reference (baseline) dummy per variable, leaving $k - 1$ columns. This has two critical benefits:

- **Invertibility of X^TX:** With perfect multicollinearity the design matrix does not have full column rank, so (X^TX) is singular and OLS coefficients cannot be uniquely estimated.

Dropping one dummy makes the matrix full-rank and the solution unique.

- **Interpretability:** The dropped level becomes the implicit baseline. Every retained dummy's coefficient measures the effect of that level *relative to the baseline*, giving clear, unambiguous interpretation — e.g., season_Winter's coefficient tells us how much more (or less) demand there is in Winter compared to Spring (the dropped level).

Note: In statsmodels, failing to use drop_first=True causes a ValueError or produces NaN coefficients because the singular matrix cannot be inverted.

Q3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

temp (actual temperature in °C) has the highest positive Pearson correlation with cnt, at approximately r ≈ 0.63. This is clearly visible in the pair-plot: the scatter of temp vs cnt shows the most distinct upward linear trend among all numeric features.

*Note: **atemp** (feels-like temperature) shows an equally high correlation (r ≈ 0.63) but is nearly perfectly collinear with temp (r ≈ 0.99 between temp and atemp). To avoid multicollinearity, atemp is dropped and only temp is retained in the model.*

Other correlations with cnt for reference: hum ≈ −0.10 (weak negative), windspeed ≈ −0.23 (weak negative), yr ≈ 0.57 (moderate positive).

Q4. How did you validate the assumptions of Linear Regression after building the model on the training set?

The five core OLS assumptions were validated using the following diagnostics after fitting the final model on the training set:

- **Linearity:** A Residuals vs Fitted Values scatter plot was examined. If the relationship is linear, residuals should be randomly scattered around the horizontal zero line with no systematic curve or pattern. The plot showed no funnel shape or curved trend, confirming approximate linearity.
- **Independence of Errors:** The Durbin-Watson statistic in the OLS summary (statsmodels) was checked. A value close to 2.0 indicates no significant autocorrelation in the residuals. Since data is daily time-series, this was an important check.
- **Homoscedasticity (Constant Variance):** The Residuals vs Fitted Values plot was inspected for a funnel/cone shape. Residuals appeared evenly spread around zero across the entire range of fitted values, suggesting that variance is constant (homoscedastic).
- **Normality of Residuals:** Two checks were applied:
  - **(a) a histogram with KDE overlay —** residuals showed an approximately bellshaped, symmetric distribution centred at zero;
  - **(b) a Q-Q plot —** the quantiles of residuals were plotted against theoretical normal quantiles, and the points lay close to the 45-degree reference line with only minor deviations in the tails.

- **No Multicollinearity:** The Variance Inflation Factor (VIF) was computed for each predictor in the final model using statsmodels' variance_inflation_factor. All VIF values were below 5, confirming that multicollinearity is not a concern. (High VIF in earlier models led to dropping atemp and iteratively removing correlated features.)

Q5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

Based on the magnitude of scaled coefficients (all continuous features were MinMax-scaled, making coefficients directly comparable) and statistical significance (p-value < 0.05), the top 3 features are:

- **temp (coefficient ≈ +0.49 to +0.55):** Temperature has the largest positive coefficient in the final model. A 1-unit increase in normalised temperature raises expected daily rentals by more than any other predictor. Warmer weather is the single biggest driver of demand — riders strongly prefer cycling in comfortable temperatures.
- **yr (coefficient ≈ +0.23 to +0.25):** The year indicator (0 = 2018, 1 = 2019) captures the yearover-year growth trend. 2019 recorded significantly more rentals than 2018, reflecting the growing popularity of the BoomBikes service. This confirms that demand is trending upward and the company should continue expanding.
- **weathersit_LightSnowRain (coefficient ≈ −0.28 to −0.32):** Days with light snow or rain carry the largest negative coefficient in the model. Adverse weather dramatically suppresses ridership — the expected daily count drops by several hundred bikes compared to a clear day. This is the most impactful negative factor and should inform fleet maintenance scheduling.

PART B - General Subjective Questions

G1. Explain the linear regression algorithm in detail.

Linear Regression is a supervised learning algorithm that models the relationship between one or more independent variables (features, X) and a continuous dependent variable (target, y) by fitting the best-possible straight line (or hyperplane) through the data.

Mathematical Formulation:

- Simple linear regression (one predictor): $y = \beta_0 + \beta_1 x + \varepsilon$
- Multiple linear regression (p predictors): $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p + \varepsilon$ where β_0 is the intercept, $\beta_1 \ldots \beta_p$ are slope coefficients, and $\varepsilon \sim N(0, \sigma^2)$ is the irreducible error (noise).

Ordinary Least Squares (OLS) — How Coefficients Are Estimated:

OLS finds the coefficient vector $\hat{\beta}$ that minimises the Residual Sum of Squares (RSS): RSS $= \Sigma_i (y_i - \hat{y}_i)^2 = \Sigma_i (y_i - X_i\beta)^2$

The closed-form (analytical) solution is:

$\hat{\beta} = (X^T X)^{-1} X^T y$

This requires $X^T X$ to be invertible (full column rank — hence the importance of removing perfect multicollinearity).

Assumptions of OLS Linear Regression:

- Linearity — the relationship between X and y is linear.
- Independence — observations (residuals) are independent of each other. • Homoscedasticity — residuals have constant variance: $Var(\varepsilon) = \sigma^2$ for all i.
- Normality of residuals — errors ε are normally distributed with mean 0.
- No perfect multicollinearity — no predictor is an exact linear combination of others.

Model Evaluation Metrics:

- **R^2** (Coefficient of Determination) — proportion of variance in y explained by the model. Range:

  0 to 1; higher is better.
- **Adjusted R^2** — penalises adding irrelevant predictors; better for comparing models with different numbers of features.
- **F-statistic / p-value** — tests whether at least one predictor is significantly related to y.
- **t-statistic / p-value per coefficient** — tests individual feature significance.
- **Residual Standard Error (RSE)** — average distance of observations from the fitted line.

G2. Explain the Anscombe's Quartet in detail.

Anscombe's Quartet, introduced by statistician Francis Anscombe in 1973, consists of four datasets that are nearly identical in standard summary statistics yet look completely different when visualised. It is one of the most powerful demonstrations in statistics, emphasising why data visualisation is essential before and after modelling.

Statistical Property	Dataset I	Dataset II	Dataset III	Dataset IV
Mean of x	9.0	9.0	9.0	9.0
Mean of y	7.50	7.50	7.50	7.50
Variance of x	11.0	11.0	11.0	11.0
Variance of y	~4.12	~4.13	~4.12	~4.12
Pearson's r	~0.816	~0.816	~0.816	~0.817
Regression line	y≈3+0.5x	y≈3+0.5x	y≈3+0.5x	y≈3+0.5x

Despite having virtually identical means, variances, correlations, and even the same regression equation, the four scatter plots reveal very different underlying structures:

- **Dataset I**: A clean linear relationship with normally distributed residuals — the textbook ideal regression scenario. Linear regression is fully appropriate here.
- **Dataset II**: A clear curved (quadratic) relationship. A linear model is the wrong choice; a polynomial or non-linear model is needed.
- **Dataset III**: Mostly linear, but one extreme outlier exerts high leverage and heavily influences the regression line, distorting coefficients and R^2.
- **Dataset IV**: All x-values are identical (x = 8) except one (x = 19). The single unusual point entirely determines the slope — without it, no regression is possible.

Key lesson: R^2, correlation, and regression coefficients can be identical for radically different data structures. Always visualise your data — pair-plots, residual plots, and Q-Q plots — before trusting numerical summaries.

G3. What is Pearson's R?

Pearson's R (Pearson Product-Moment Correlation Coefficient) is a measure of the strength and direction of the linear association between two continuous variables X and Y. It is the most widely used measure of correlation in statistics.

Formula:

$r = Cov(X,Y) / (\sigma_X \cdot \sigma_Y) = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2]}$ **Interpretation:**

- **r = +1**: perfect positive linear relationship — as X increases, Y increases proportionally.
- **r = −1**: perfect negative linear relationship — as X increases, Y decreases proportionally.
- **r = 0**: no linear association (non-linear relationships may still exist — see Anscombe's Quartet).
- **|r| > 0.7** is generally considered strong; 0.4 - 0.7 moderate; < 0.4 weak.

Key Properties:

- Dimensionless and scale-invariant: multiplying X or Y by any constant does not change r.
- Symmetric: r(X, Y) = r(Y, X).
- Sensitive to outliers: a single influential point can shift r dramatically.
- Only measures linear dependence: r = 0 does not imply independence.

In this project: temp had r ≈ 0.63 with cnt (strong positive — the top numeric predictor). hum had r ≈ −0.10 (very weak negative) and windspeed r ≈ −0.23 (weak negative).

G4. What is scaling? Why is scaling performed? What is the difference between normalised scaling and standardised scaling?

Scaling is the process of transforming the numeric values of features to a common range or distribution. Raw features often span vastly different magnitudes and units (e.g., temperature in °C vs. wind speed in m/s), which can distort certain algorithms.

Why is scaling performed?

- Gradient-based optimisers (e.g., gradient descent) converge much faster when all features are on a similar scale — otherwise the cost surface is elongated and updates are slow.
- Regularisation methods (Ridge, Lasso) penalise coefficient magnitude. Without scaling, features with larger raw values get unfairly penalised relative to smaller-magnitude features.
- Distance-based algorithms (KNN, SVM, K-Means, PCA) are dominated by high-magnitude features without scaling.
- Makes OLS coefficients directly comparable for judging feature importance (as used in this project to identify the top 3 features).

Normalisation (Min-Max Scaling): $x' = (x - x_{min}) / (x_{max} - x_{min})$

Compresses all values to the interval [0, 1]. Preserves the original distribution shape. Sensitive to outliers — one extreme value compresses all other values toward 0 or 1. Used in this project for temp, hum, windspeed.

Standardisation (Z-score Scaling): $x' = (x - \mu) / \sigma$

Centres data at mean 0 with standard deviation 1. More robust to outliers than min-max scaling. Does not bound values to a fixed range — output can be any real number. Preferred when data is approximately Gaussian or when using regularised regression (Ridge, Lasso).

Property	**Normalisation (Min-Max)**	**Standardisation (Z-score)**
Output range	[0, 1]	Unbounded (any real number)
Formula	(x − min)/(max − min)	(x − μ)/σ
Outlier sensitivity	High	Lower
Preserves shape	Yes	Yes
When to use	Bounded features, no severe outliers	Gaussian data, regularisation

G5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

The Variance Inflation Factor (VIF) for predictor X_j is defined as: $VIF_j = 1 / (1 - R^2_j)$

where R^2_j is the R^2 obtained by regressing X_j on all other predictors in the model. It measures how much of X_j's variance is explained by the other features.

Why VIF becomes infinite:

When $R^2_j = 1$ — i.e., predictor X_j can be perfectly predicted from the other predictors — the denominator becomes $(1 - 1) = 0$, making VIF → ∞. This is perfect multicollinearity: X_j is an exact linear combination of one or more other features.

Common causes of infinite VIF in practice:

- **Dummy variable trap**: Including all k dummies for a k-level categorical variable without drop_first=True creates a column that is an exact linear combination of the others (they always sum to 1). This is the most frequent cause.
- **Derived / redundant columns**: Adding a feature that is algebraically derived from others already in the model — e.g., including 'Total_Rentals', 'Casual', and 'Registered' simultaneously, where Total = Casual + Registered.
- **Duplicate columns**: The same feature accidentally included twice under different names.
- **Constant columns**: A feature with zero variance (same value in every row) cannot be standardised and causes a singular matrix.

Resolution:

Use drop_first=True when creating dummies; drop derived or duplicate columns; check for zerovariance features before modelling. In this project, dropping casual, registered, and atemp — and using drop_first=True — eliminated all cases of infinite or very high VIF.

G6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

A Quantile-Quantile (Q-Q) plot is a graphical tool that assesses whether a dataset follows a specified theoretical distribution — most commonly the standard normal distribution. It plots the empirical (observed) quantiles of a sample on the y-axis against the corresponding theoretical quantiles on the x-axis.

How to construct and read a Q-Q plot:

- Sort the data values and compute their empirical quantiles (percentile ranks).
- Compute the corresponding theoretical quantiles from the standard normal distribution for the same percentile ranks.
- Plot (theoretical quantile, empirical quantile) for each data point.
- **Points lie on the diagonal y = x line**: The data closely follows the theoretical distribution → normality assumption holds.
- **S-shaped (sigmoid) curve**: The distribution has heavier tails (leptokurtic) or lighter tails (platykurtic) than normal.
- **Points curve away at one or both ends**: The distribution is skewed (right-skewed if the upper tail rises above the line; left-skewed if it falls below).
- **Individual points far off the line**: Possible outliers or influential observations that warrant further investigation.

Importance in Linear Regression:

One of the core OLS assumptions is that residuals are normally distributed: $\varepsilon \sim N(0, \sigma^2)$. Violating this assumption has real consequences:

- **Unreliable hypothesis tests**: t-tests and F-tests rely on normality; non-normal residuals produce incorrect p-values, leading to wrong conclusions about which features are significant.
- **Invalid confidence intervals**: Prediction intervals are too wide or too narrow if residuals are not normal.
- **Signals model problems**: Heavy-tailed residuals often indicate outliers or the need for a transformation (e.g., log(y)); skewed residuals suggest a non-linear relationship or a missing predictor.

Advantage over histogram: A Q-Q plot is more sensitive to tail behaviour and deviations from normality than a histogram, especially with small samples where a histogram's shape is heavily influenced by bin width.

In this project: The Q-Q plot of training residuals showed points lying close to the 45-degree reference line with only minor deviations in the extreme tails, confirming that the normality assumption is approximately satisfied and OLS inference is valid.